

Fish consumption, insulin sensitivity and beta-cell function in the Insulin Resistance Atherosclerosis Study (IRAS)

Background and aims Previous research on the association between fish consumption and incident type 2 diabetes has been inconclusive. In addition, few studies have investigated how fish consumption may be related to the metabolic abnormalities underlying diabetes. Therefore, we examined the association of fish consumption with measures of insulin sensitivity and beta-cell function in a multi-ethnic population. **Methods and results** We examined the cross-sectional association between fish consumption and measures of insulin sensitivity and secretion in 951 non-diabetic participants in the Insulin Resistance Atherosclerosis Study (IRAS). Fish consumption, categorized as <2 vs. ≥ 2 portions/week, was measured using a validated food frequency questionnaire. Insulin sensitivity (SI) and acute insulin response (AIR) were determined from frequently sampled intravenous glucose tolerance tests. Higher fish consumption was independently associated with lower SI-adjusted AIR ($\beta = -0.13$ [$-0.25, -0.016$], $p = 0.03$, comparing ≥ 2 vs. <2 portions/week). Fish consumption was positively associated with intact and split proinsulin/C-peptide ratios, however, these associations were confounded by ethnicity (multivariable-adjusted $\beta = 0.073$ [$-0.014, 0.16$] for intact proinsulin/C-peptide ratio, $\beta = 0.031$ [$-0.065, 0.13$] for split proinsulin/C-peptide ratio). We also observed a significant positive association between fish consumption and fasting blood glucose (multivariable-adjusted $\beta = 2.27$ [$0.68, 3.86$], $p = 0.005$). We found no association between fish consumption and SI (multivariable-adjusted $\beta = -0.015$ [$-0.083, 0.053$]) or fasting insulin (multivariable-adjusted $\beta = 0.016$ [$-0.066, 0.10$]). **Conclusions** Fish consumption was not associated with measures of insulin sensitivity in the multi-ethnic IRAS cohort. However, higher fish consumption may be associated with pancreatic beta-cell dysfunction.